

Michail Mitsakis *Materials Scientist*

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📄 Summary

Materials scientist focused on electrochemical synthesis for industrial decarbonization and AI-driven materials discovery. Hands-on with Bayesian optimization and building agentic tools for R&D workflows. Consulting on EU energy projects has shown me how scientific innovation translates into actionable insights and real-world impact.

📁 Professional Experience

Junior Business Consultant, Energy, WMG Sustainable Innovation

2024 – present | Greece

- **EU R&I proposals**: Co-authored **15** proposals (5 Horizon Europe, 8 CEF, 2 Digital Europe) on maritime transport, liquid hydrogen, carbon-capture membranes and data spaces, translating early-stage innovation concepts into actionable workplans. Built a **>€10M** lump-sum budget (rates benchmarked on EC data, depreciation, cost justification).
- **Feasibility analyses**: Delivered techno-economic feasibility studies, a CEF cost-benefit analysis and market mapping reports. Preparing viable business models and IPR-strategy considerations for project portfolios valued at **>€10M**, such as hydrogen retrofit carrier and liquid-hydrogen cargo containment Horizon projects.
- **Project management**: Executed reporting for Clean Hydrogen JU flagships EPHYRA (50 MW electrolyser) and TRIERES (Greece's first Hydrogen Valley), **>€35M** combined.
- **Data spaces & knowledge graphs**: Co-authored **2** deliverables for data space UNDERPIN on Digital Product Passports. Drafting a paper mapping software properties and dependencies to semantic ontologies for knowledge-graph (KG) interoperability.

Personnel Officer, Hellenic Air Force Academy (mandatory military service)

2023 – 2024 | Greece

🎓 Education

M.Sc. in Engineering - Physics and Nanotechnology, Technical University of Denmark

2021 – 2023 | Denmark

Thesis (DTU Energy): "Machine learning-driven optimization of electrodeposited Ni-W electrocatalysts for H₂ evolution in alkaline electrolysis." Built a multi-objective Bayesian-optimization loop over 4 electrodeposition parameters, optimizing catalyst activity and stability through minimizing the electrode overpotential and its change after stress testing; characterized ~30 electrodes.

B.Sc. in Physics, National and Kapodistrian University of Athens

2015 – 2020 | Greece

Thesis (NCSR "Demokritos"): "CO₂ sorption and permeability in supported single-walled carbon-nanotube membranes."

📁 Projects

catalyst-kg-agent

Knowledge-graph-grounded multi-agent system for HER catalyst screening under a fixed budget. It decides the cheapest next step (KG lookup, MACE/CGCNN surrogate or expensive verification) using Materials Project data.

notion-second-brain

A fully local RAG agent answering queries on my Notion notes; OCR ingestion, hybrid retrieval, cross-encoder reranking, Qdrant DB and a rubric-based eval harness.

Materials design with ML & AI, DTU

Gaussian-process prediction of elastic moduli for 3,858 2D materials. Team ranked 3rd of 14 in course's Kaggle competition.

Engineering covalently bonded 2D layered materials by self-intercalation for the HER, DTU

DFT study of 5 TaS₂ (TMD) bilayers, 0–100% intercalated, for HER. S-on-top was the strongest H-binding site; 75% intercalation as a stability–activity compromise: H binding –2.14 eV, near-zero H₂ binding (spontaneous desorption), moderate stability.

3D Pyrolytic Carbon-Gold Electrodes, DTU Nanolab

Team-based cleanroom work in operating multiple advanced experimental setups (spin-coating, baking, development, lithography, pyrolysis, deposition, optical microscopy, SEM, stylus profilometry), towards maximizing the energy efficiency of electrodes for energy harvesting applications.

CO₂ electrolysis to formate, DTU Physics

Investigation of formate utilization in improving fuel cell energy efficiency. Measured crossover across multiple membranes and electrolytes in H-cell and MEA configurations (CV, HPLC, GC).

X-Tech Entrepreneurship, DTU

R&D Expert in startup developing an electrochemical sensor of micro-organic particles towards minimizing crop yield losses. Experience with cleanroom work, market analysis, progress reports, strict deadlines and deliverables.

Green fuels & Power-to-X, DTU

Feasibility study on the optimized production of e-methanol as maritime fuel. Aspen Plus used as modelling software.

Publications & Awards

Roy, A., ..., Mitsakis, M., et al. "From Knowledge to Action: Outcomes of the 2025 Large Language Model (LLM) Hackathon for Applications in Materials Science and Chemistry." arXiv:2605.03205 (2026)

CaMEL-RAG (5-person team): a citation-grounded RAG over Open Catalyst adsorption energies that returned the exact DFT value for all 500 in-index test queries.

Baird, S., ..., Mitsakis, M., et al. "Bayesian Optimization Hackathon for Chemistry and Materials." ChemRxiv (2025)., doi/10.26434/chemrxiv-2025-dzh5z

3rd place (4-person team). BayBE screening of corrosion inhibitors across 6 aluminium alloys, comparing 4 molecular encodings. Transfer learning reached ~96% inhibition efficiency in 12 experiments, vs ~89% after 25 from scratch.

Skills

- **Core:** Cross-functional collaboration, Project & conflict management, Systems thinking, Adaptability, Technical writing
- **Materials Informatics:** Python, PyTorch, Bayesian optimization (BoTorch/Ax, BayBE, Dragonfly), DFT (ASE, GPAW), MLIPs
- **Lab Techniques:** Electrodeposition, CP, CV, EIS, HPLC, GC, Cleanroom nanofabrication (spin-coating, baking, development, lithography, pyrolysis, CVD, optical microscopy, SEM, stylus profilometry)
- **Agentic AI:** Local RAG, LLM-as-a-Judge, MCP, Observability (Arize, MLFlow)

Certificates

- Arize AI Agents Mastery (2026)
- UC Berkeley Agentic AI MOOC (2026)
- DataTalksClub LLM Zoomcamp (2025)
- Hugging Face MCP Course (2025)
- Climate Change AI Summer School (2023, 2024)
- IFP School Hydrogen Production (2024)

Courses

AI for Experimental Chemistry (2026), Washington University, CHEM 5080

Chemical Data Science (2026), Rutgers University - The Sun Lab, CHEM 487-542

AI Agents for Science (2025), UChicago, CMSC 35370

Intro to Deep Learning (2025), MIT, 6.S191

AI in the Sciences and Engineering (2024), CAMLab ETH Zürich

AI for Chemistry (2023), EPFL - Schwaller Group, CH-457

Deep Generative Models: Probabilistic foundations and learning algorithms (2023), Cornell, Stanford, UCLA, Caltech; joint course, CS 236

Practical Deep Learning for Coders (Part 1), fastai

Languages

Greek • **English** — C2 & IELTS (2021) • **French** — B2 • **German** — B2

Interests

Calisthenics · Reading (fantasy, sci-fi, philosophy, science, history) · Writing · Philosophy of Mind · Ontologies of Quantum Mechanics · Music (rock, punk, psychedelic, funk, jazz, blues, indie)